Mass-Spring-Damper (MSD) System

Using

Simulink and MATLAB

Summary:

This project involves the modeling and simulation of a Mass-Spring-Damper System using MATLAB and Simulink. The mathematical equation of the system was implemented in Simulink using standard simulation blocks. MATLAB was used to define the system parameters, including mass, damping coefficient, and spring constant. The simulation generated graphical outputs to study the system's dynamic behavior. This project demonstrates the use of MATLAB and Simulink for modeling and analyzing a mechanical system.

Fundamental Concepts:

The Mass-Spring-Damper (MSD) System is one of the most fundamental mechanical systems used in engineering to model the motion of objects subjected to mass (inertia), spring (elasticity), and damping (energy dissipation). It is widely used in Mechanical Engineering.

A mass-spring-damper system describes how an object moves when external forces act upon it while being connected to a spring and a damper. The spring tries to restore the object to its equilibrium position, while the damper resists motion by dissipating energy, usually in the form of heat.

System Under Consideration

Quarter-Car Suspension Model:

The Quarter-car Model is a simplified representation of a vehicle's suspension system used to study its vertical motion and vibration. Instead of modeling the entire vehicle, only one-quarter of the vehicle is considered because all four corners of the suspension behave in a similar manner under normal driving conditions. This simplification reduces the complexity of the analysis while still providing an accurate understanding of the suspension's dynamic behavior.

Physically, the model consists of a mass, a spring, and a damper. The mass represents one-fourth of the vehicle body (sprung mass), the spring represents the suspension spring that supports the vehicle's weight, and the damper represents the shock absorber that reduces unwanted vibrations. The road surface is treated as a moving base or input, which excites the suspension system when the vehicle passes over bumps, potholes, or uneven roads.

Mathematical Equation:

$$\ddot{x} = -\frac{c}{m}x - \frac{b}{m}\dot{x} + \frac{b}{m}\dot{z} + \frac{c}{m}z \text{......(Equation 1)}$$

OR

$$\ddot{x} = -\frac{c}{m}(x - z) - \frac{b}{m}(\dot{x} - \dot{z}) \text{.....(Equation 2)}$$

Visual Representation of Quarter-Car Model

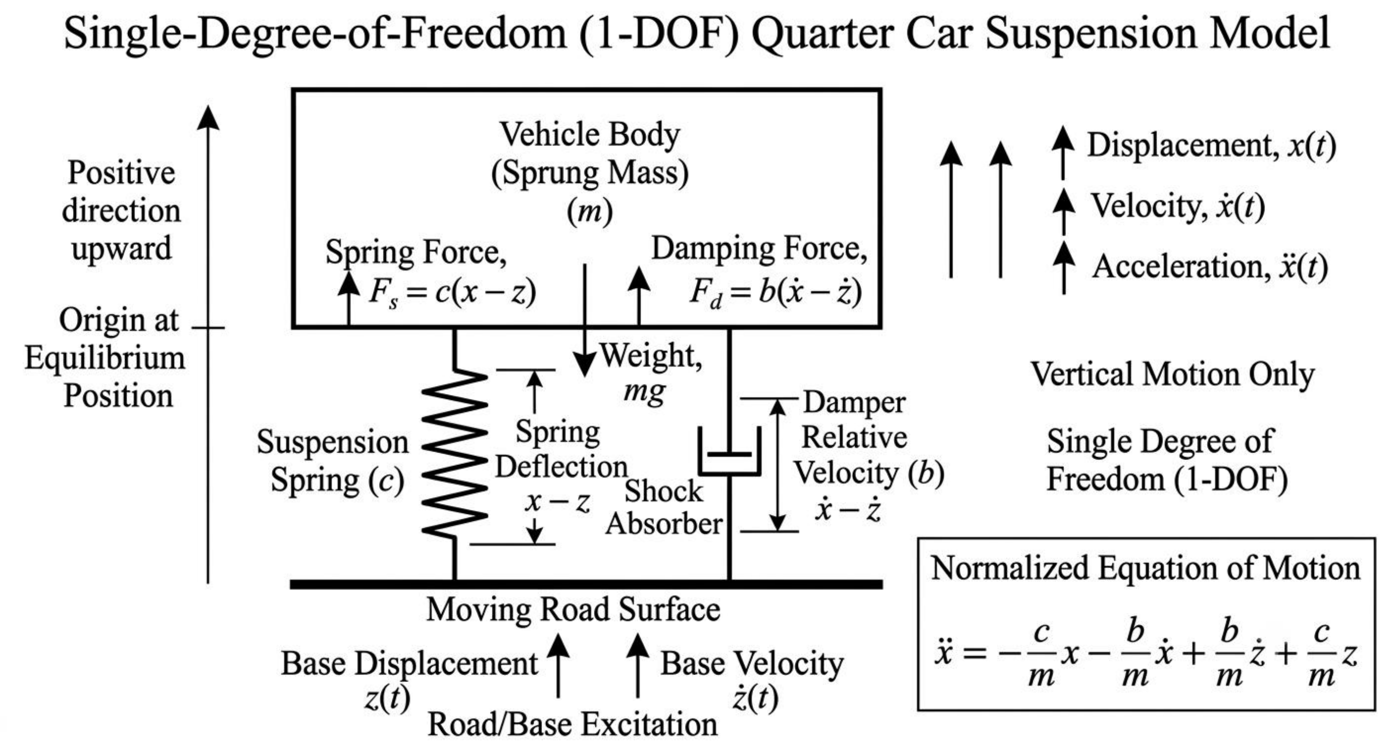
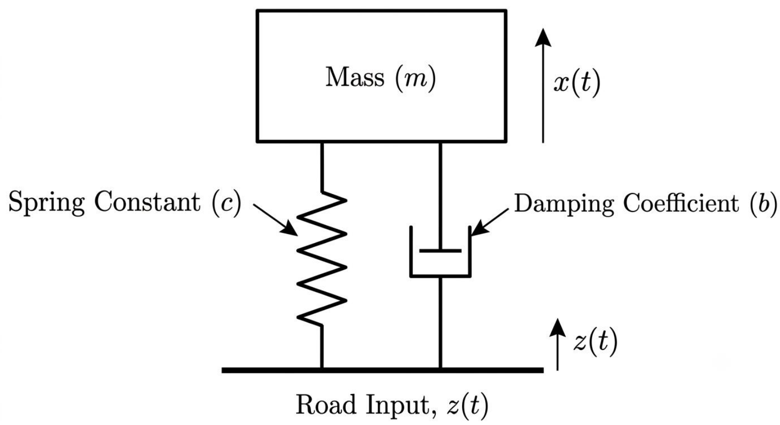


Figure (i)

Representation of Parameters:

Sno	Symbol	Parameter	Units	Explanation
1.	m	Mass (sprung)	Kilograms	$-\frac{c}{m}x$ Represents restoring force of spring, which opposes the displacement of mass.
2.	b	Damping coefficient	N.s/m	
3.	c	Spring constant (stiffness)	N/m	$-\frac{b}{m}\dot{x}$ Represents the damping force, which resists the motion of the mass.
4.	x	Displacement of the mass	meters	
5.	$\dot{x}$	Velocity of the mass	m/s	$+\frac{b}{m}\dot{z}$ Represents effect of the base (input) velocity transmitted through the damper.
6.	$\ddot{x}$	Acceleration of the mass	m/s^2	
7.	z	Base displacement	meters	$+\frac{c}{m}z$ Represents the effect of base (input) displacement transmitted through spring.
8.	$\dot{z}$	Base velocity	m/s	



(A simplified figure of the Quarter-car model)

Figure (ii)

Introduction to MATLAB:

MATLAB (Matrix Laboratory) is a high-level programming and numerical computing software developed by MathWorks. It is widely used in engineering, science, and mathematics for solving complex mathematical problems, analyzing data, developing algorithms, and creating simulations. MATLAB provides a user-friendly environment where mathematical models can be implemented using scripts and visualized through graphs and plots.

In this project, MATLAB was used to define the system parameters of the Quarter-car model, including the mass (m), damping coefficient (b), and spring constant (c). These parameters were then linked to the Simulink model to perform the simulation and analyze the dynamic response of the system. The integration of MATLAB with Simulink allows mathematical models to be simulated efficiently without performing lengthy manual calculations, making it a powerful tool for mechanical engineering applications.

```
% Project Quantities

m = 300         % mass- kilograms
b = 30          % Damping
c = 100       % Spring constant
```

Paramters for model - I

```
% Project Quantities

m = 300         % mass- kilograms
b = 300          % Damping
c = 10000       % Spring constant
```

Parameters for Model - II

Sno	Parameters	Symbol	Unit	Values
1.	Mass	m	Kilograms (kg)	300
2.	Damping coefficient	b	N.s/m	30 and 300
3.	Spring constant	c	N/m	100 and 10,000
4.	Simulation time	---	---	20 seconds

Introduction to the Simulink:

Simulation is a computational technique used to represent and analyze the behavior of real-world systems through mathematical models. It enables engineers to study the response of a system under various operating conditions without conducting physical experiments, thereby reducing cost, time, and risk. In this project, MATLAB Simulink is utilized as the primary simulation platform due to its graphical block-diagram environment, which allows mathematical equations and dynamic systems to be modeled, simulated, and analyzed efficiently. Simulink also provides visualization tools that help engineers evaluate system performance and validate theoretical concepts.

In this project, simulation is performed on a quarter-car suspension model to investigate its dynamic response to road surface disturbances. The mathematical equations governing the suspension system are implemented in Simulink using the specified parameters, including the sprung mass, suspension spring stiffness, damping coefficient, and external road input. The simulation generates the displacement, velocity, and acceleration responses of the vehicle body, allowing the performance of the suspension system to be analyzed in terms of ride comfort and vehicle stability under different operating conditions.

Simulink Working

The quarter-car suspension system was implemented using two different Simulink models.

Model - I:

Quarter-car Suspension Model

Vehicle Velocity and Displacement Response

Procedure for Model - I:

The first Simulink model was developed to simulate the dynamic behaviour of a quarter-car suspension system subjected to a sinusoidal road disturbance. A Sine Wave block was used to generate the road displacement input (z), while a Derivative block calculated the corresponding road velocity $(\dot{z})$. These two signals were multiplied by the gain blocks c/m and b/m, representing the spring stiffness and damping effects of the suspension system. The resulting signals were added using a Sum block to form the external excitation acting on the vehicle. An Add block then combined the excitation with the negative feedback terms corresponding to the spring and damping forces generated by the vehicle body. Two cascaded Integrator (1/s) blocks converted the calculated acceleration into velocity and subsequently into displacement. Finally, a Mux block combined the velocity and displacement outputs, which were displayed simultaneously on a Scope block for analysis.

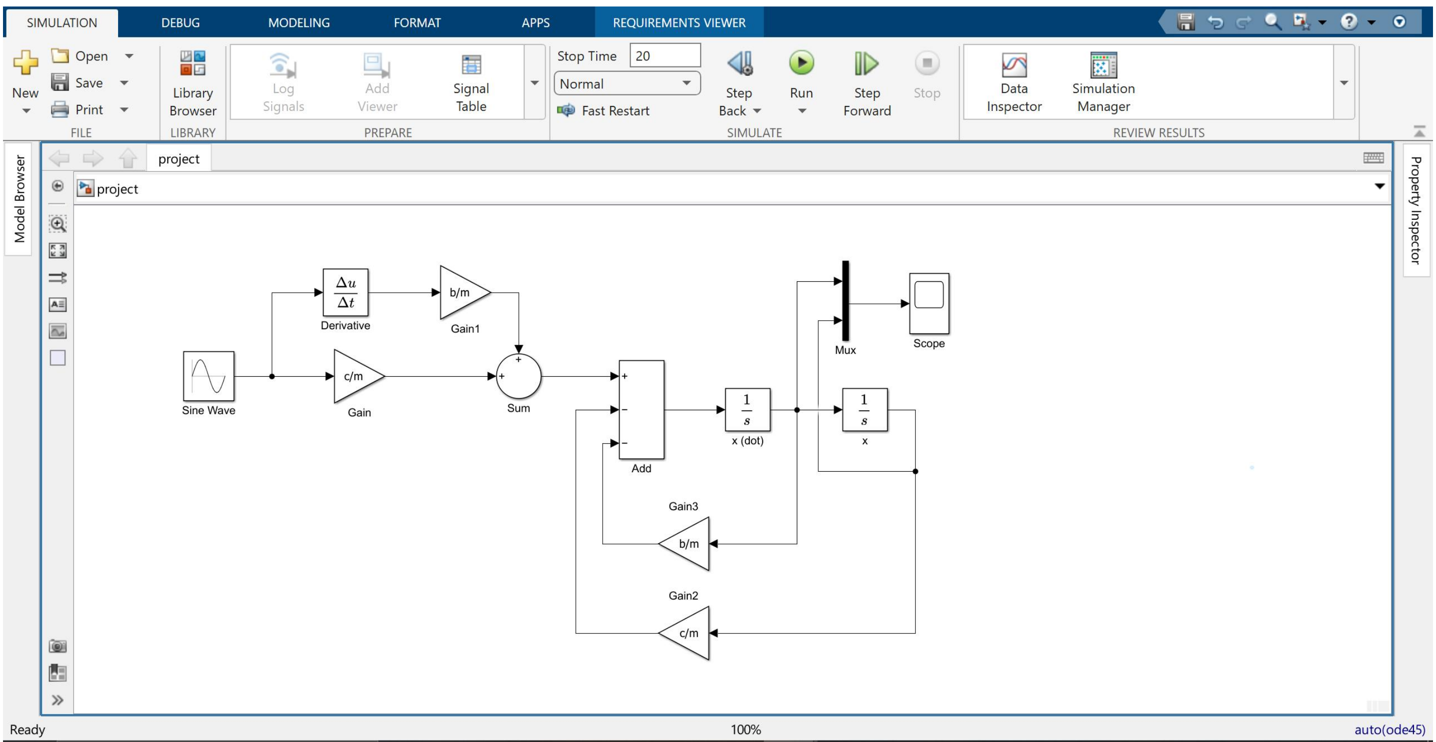


Figure (iii) - Simulink Model (i)

The model operates according to the governing differential equation of the quarter-car suspension system. The sinusoidal road profile continuously excites the suspension, producing both displacement and velocity inputs. These inputs generate forces through the spring and damper, which are balanced against the restoring forces produced by the vehicle suspension. The Add block calculates the net acceleration of the sprung mass, which is then integrated to obtain the vehicle velocity and displacement. The feedback loops continuously update the restoring spring force and damping force, enabling the model to simulate the dynamic interaction between the road surface and the vehicle body. This closed-loop configuration accurately represents the vibration characteristics of the suspension system under harmonic road excitation.

Scope Result of Model - I :

The Scope graph shows that the simulation was carried out for 20 seconds. Both the velocity and displacement responses exhibit periodic oscillations because the suspension system is excited by a sinusoidal road input. During the initial 1–2 seconds, the system undergoes a transient response while adjusting to the external disturbance. After this period, the oscillations become regular, indicating that the system has reached steady-state operation. The vehicle displacement varies approximately between −1 m and +1 m, resulting in a peak-to-peak displacement of approximately 2 m. The vehicle velocity reaches a maximum value of approximately ±1 m/s. The amplitudes remain nearly constant throughout the simulation, demonstrating that the damping coefficient effectively prevents the oscillations from increasing with time. The response remains bounded, confirming that the suspension system is dynamically stable.

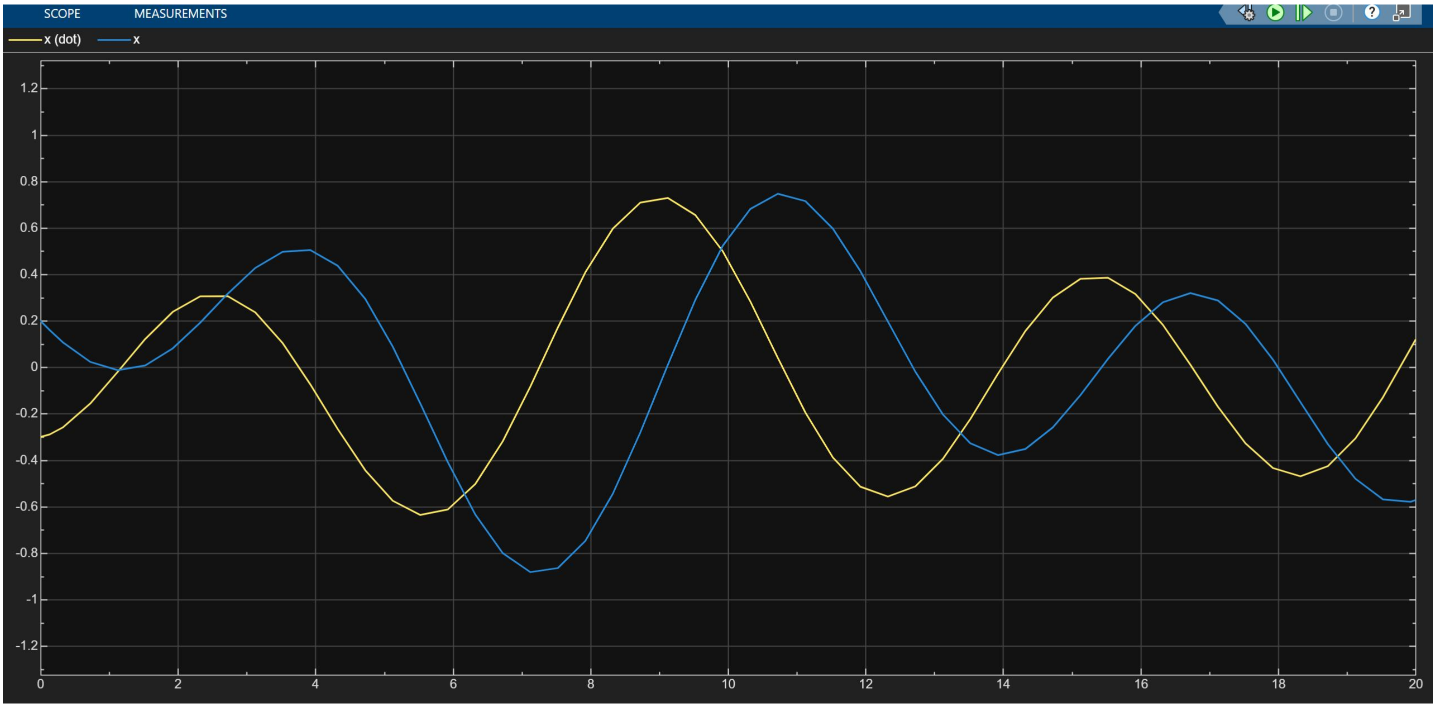


Figure (iv) - Result of model (i)

Quarter-car Suspension Model

Road Input and Vehicle Displacement Comparison

Procedure for Model -II:

The second Simulink model uses the same quarter-car suspension configuration as the first model, with the primary difference being the selection of output signals. The road displacement generated by the Sine Wave block is connected directly to one input of the Mux block, while the vehicle displacement obtained from the second integrator is connected to the second input. The remainder of the model, including the Derivative block, gain blocks, summation blocks, feedback loops, and integrators, remains unchanged. This arrangement allows the Scope to display the road excitation and the resulting vehicle displacement simultaneously for direct comparison.

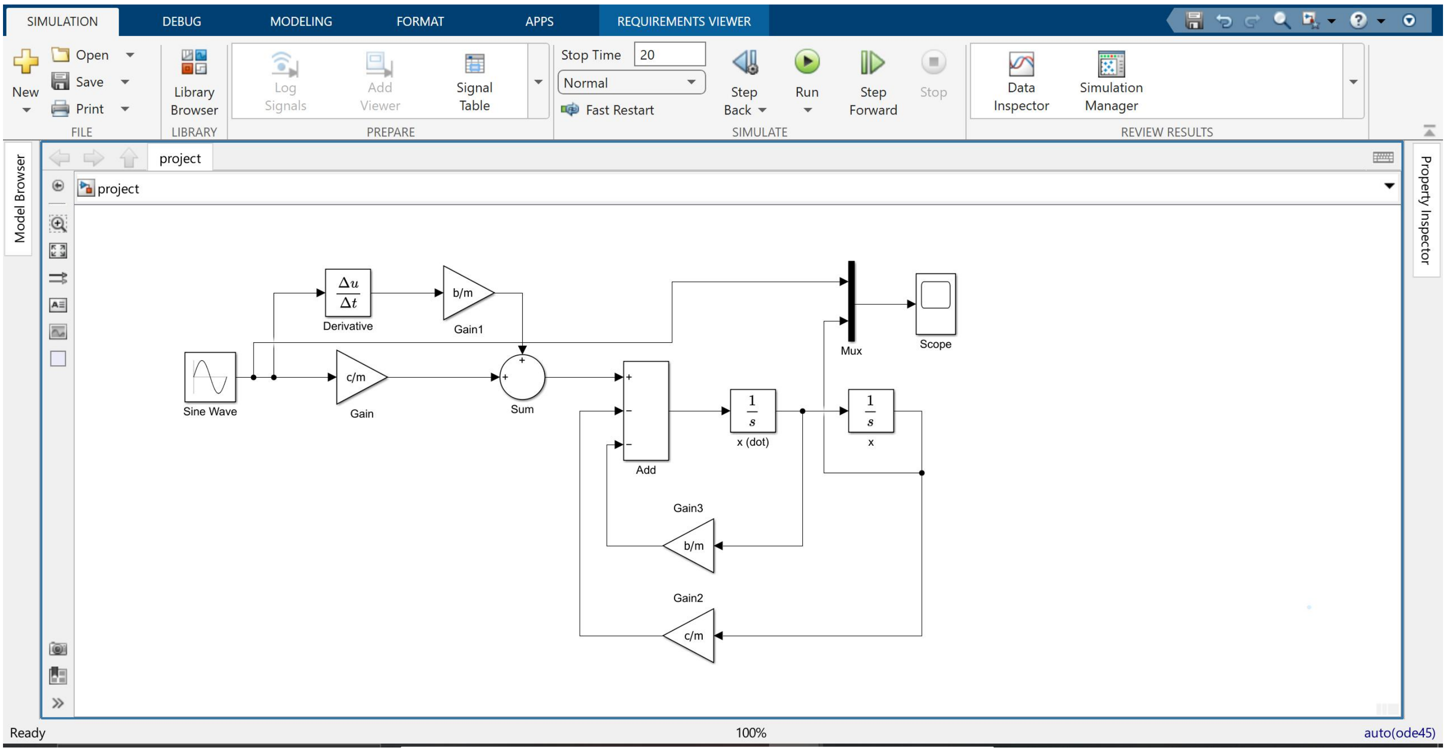


Figure (v) - Simulink Model (ii)

The operating principle of this model is identical to that of the previous configuration. The sinusoidal road profile acts as the excitation source, and the suspension system computes the dynamic response of the sprung mass using the governing mathematical equation. Instead of displaying velocity, the model compares the input road profile with the output vehicle displacement. This comparison demonstrates how effectively the suspension system isolates the vehicle body from road irregularities. By continuously balancing spring and damping forces against the external disturbance, the suspension reduces the transmission of vibrations to the vehicle body while maintaining dynamic equilibrium.

Scope Result of Model - II :

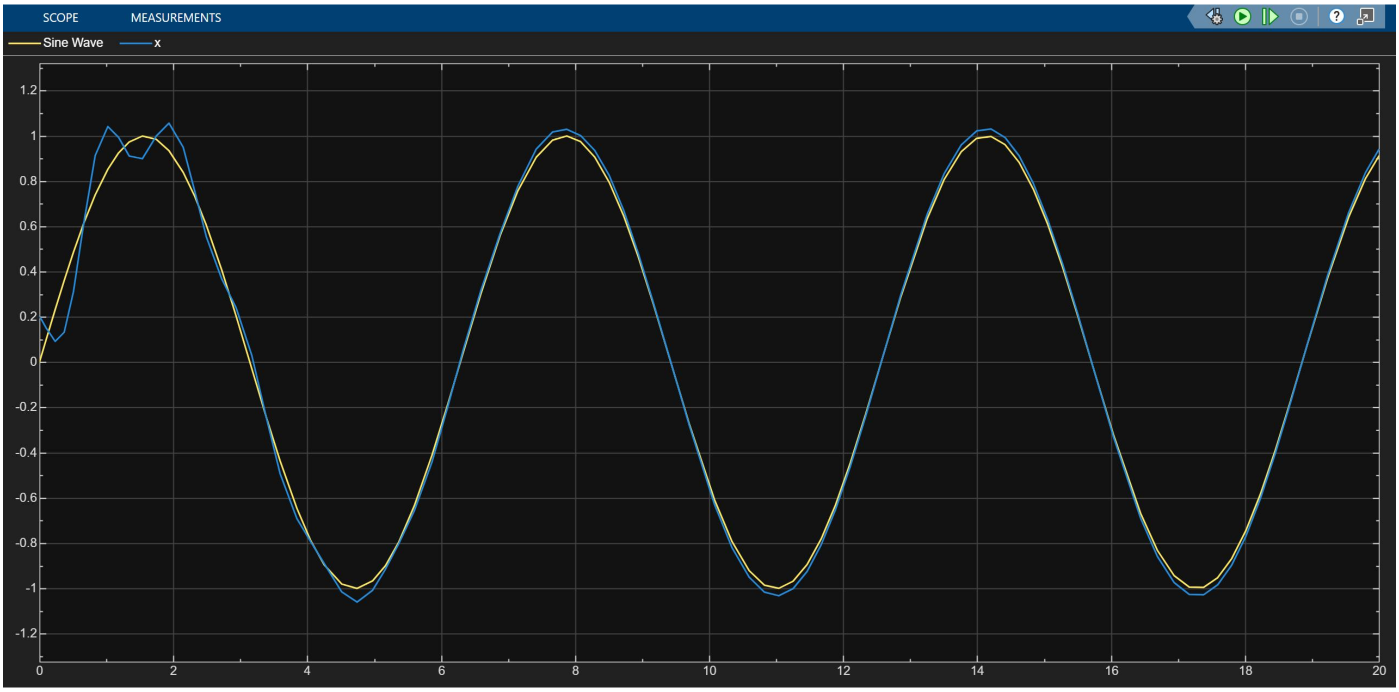


Figure (vi) - Result of Model (ii)

The second simulation also runs for 20 seconds. The road displacement and vehicle displacement follow a sinusoidal pattern with approximately the same frequency. The vehicle displacement shows a small phase delay compared to the road input, indicating that the suspension requires a short time to react to changes in the road surface. The maximum displacement of both signals is approximately ±1 m, while the oscillation period is close to 6.3 seconds, corresponding to one complete sine-wave cycle. After the initial transient response, which lasts approximately 1–2 seconds, both signals become synchronized and continue oscillating with a nearly constant amplitude. This behaviour demonstrates that the suspension successfully follows the road excitation while maintaining stable vehicle motion.

Conclusion

This project successfully developed and simulated a quarter-car suspension system using MATLAB Simulink. The main objective was to understand the dynamic behaviour of a vehicle suspension by implementing its mathematical model in a simulation environment. The governing differential equation was converted into a block diagram using Simulink components such as the Sine Wave, Derivative, Gain, Sum, Add, Integrator, Mux, and Scope blocks. Each block represented a specific mathematical or physical element of the suspension system, making it easier to visualize and analyze the vehicle's response to road disturbances.

Two different Simulink configurations were created during the project. In the first model, the vehicle's velocity and displacement responses were observed to study the motion of the sprung mass. In the second model, the road input was compared directly with the vehicle displacement to evaluate how effectively the suspension isolated the vehicle body from road vibrations. The simulation results showed that the suspension system remained stable throughout the simulation and responded smoothly to the sinusoidal road excitation. The displacement and velocity graphs exhibited continuous oscillations, which are expected due to the harmonic nature of the road input. The damping and spring stiffness prevented excessive vibrations and maintained a controlled system response.

The project also demonstrated the importance of simulation in engineering design. MATLAB Simulink provided a simple and efficient platform for modelling, testing, and analysing the suspension system without the need for a physical prototype. It allowed the mathematical equations to be implemented accurately and the system performance to be evaluated under different conditions. Overall, the project improved the understanding of vehicle suspension dynamics, feedback systems, and numerical simulation techniques. The developed model can serve as a foundation for more advanced suspension studies, such as active or semi-active suspension systems, different road profiles, and parameter optimization to further enhance ride comfort, vehicle stability, and overall suspension performance.

References:

* Mathworks: simulink
* Mathworks: MATLAB
* Gillespie, T. D. Fundamentals of Vehicle Dynamics.
* Rao, S. S. Mechanical Vibrations.